

Day-ahead load forecasting for heat-pump households: one global model, plus a PV/non-PV split

Bringing the Heat (E.ON challenge), Levels 0 and 1 | Team: Anita Aghazadeh

Approach

Data. Hourly day-ahead forecasts of household net consumption. We summed the 15-minute readings to hourly values (only complete hours kept; 406 households with a total-load signal, 6.9M household-hours) and joined the hourly weather of each household's nearest station. Everything is in three short scripts (prepare, features, models).

No leakage. The forecast origin is the end of day D-2, stricter than the real gate closure at noon on D-1, so every consumption feature for day D is at least 48 h old: same-hour lags (48/72/96/168 h), mean and std of the same hour over 7 days, recent level (last day, 7-day, 28-day means), local calendar features, and weather at the target hour (temperature, humidity, sunshine, wind, rain, heating degrees). The split is strictly chronological.

Level 0. One global LightGBM model for all households (early stopping on the last 10% of the training period), compared with three naive rules: same hour last week, same hour two days ago, and the 7-day same-hour mean.

Level 1. PV owners are found from their consumption pattern (option b). 131 households carry a PV flag and 111 a non-PV flag; for the unflagged ones a random forest uses summer midday share of load, midday-to-night and midday-to-evening ratios, summer/winter ratio and the correlation of midday share with temperature, all computed on the training period only. Cross-validated on the flagged households it reaches AUC 0.81 and 75% accuracy (AUC 0.89 / 82% with a longer training window). We compare three setups: the global model, the global model with a PV-flag input, and separate PV and non-PV models.

Experimental design

Train on all data before 2023-01-01, test on every hour of 2023 (403 households, 3.2M forecasts), which covers all seasons. Two extra checks gave consistent results: the last 20% of the data (Oct 2023 to Feb 2024) and Apr to Sep 2023. Metric: WAPE = total absolute error / total actual load. We report it per household-hour and for the summed portfolio per hour and per day, since the portfolio is what is bought on the day-ahead market.

Results (2023 test, portfolio WAPE per hour, lower is better)

Model	All	PV	non-PV
Same hour last week	20.2%	23.6%	19.2%
Mean of same hour, last 7 days (best naive)	15.9%	18.5%	15.1%
LightGBM, global (Level 0)	8.5%	11.2%	8.2%
LightGBM, global + PV flag	8.4%	10.9%	8.1%
LightGBM, separate PV / non-PV models	8.3%	11.0%	7.9%

Accuracy. The global model roughly halves the day-ahead error of the best naive rule, in every month of the year, and the daily portfolio error falls from 13.4% to 6.2%. Error per single household-hour stays high (43% vs 46%), as expected for individual homes; aggregation is what makes the forecast usable for procurement. Errors peak in spring and October (about 10%), when heating switches on and off, and are lowest in June and July (about 6%).

PV vs non-PV. PV households are consistently harder to forecast, by about 3 percentage points (11.2% vs 8.2%), and by more on the households with a known PV flag (13.5%). Separate PV and non-PV models gain only 0.2 to 0.3 points over the global model, which is within the roughly 0.4-point run-to-run variation we saw between training runs, so we do not claim a benefit. The practical value of identifying PV households is risk sizing: they need larger safety margins on the day-ahead bid.

Limitations

The data holds net consumption only, so we forecast grid draw, not PV production. Weather is the observed weather, a stand-in for a forecast, so real errors would be somewhat higher (weather ranks far below recent consumption in feature importance). Households without a week of history are excluded from scoring, and 29 households with no training history default to non-PV. We did not attempt Level 2 (cost metrics) or Level 3 (uncertainty).

Reproduce

python 01_prepare.py <repo> work; python 02_features.py <repo> work; python 03_models.py work results 2023-01-01T00:00 2023-01-01T00:00 2024-01-01T00:00 year2023 (about 10 minutes, 2 cores, 6 GB RAM). Python with polars, pandas, scikit-learn and lightgbm.